[V8 Patched Edition]

Tying the Data Knot

A State-of-the-Art Machine Learning Pipeline for Digital Romance

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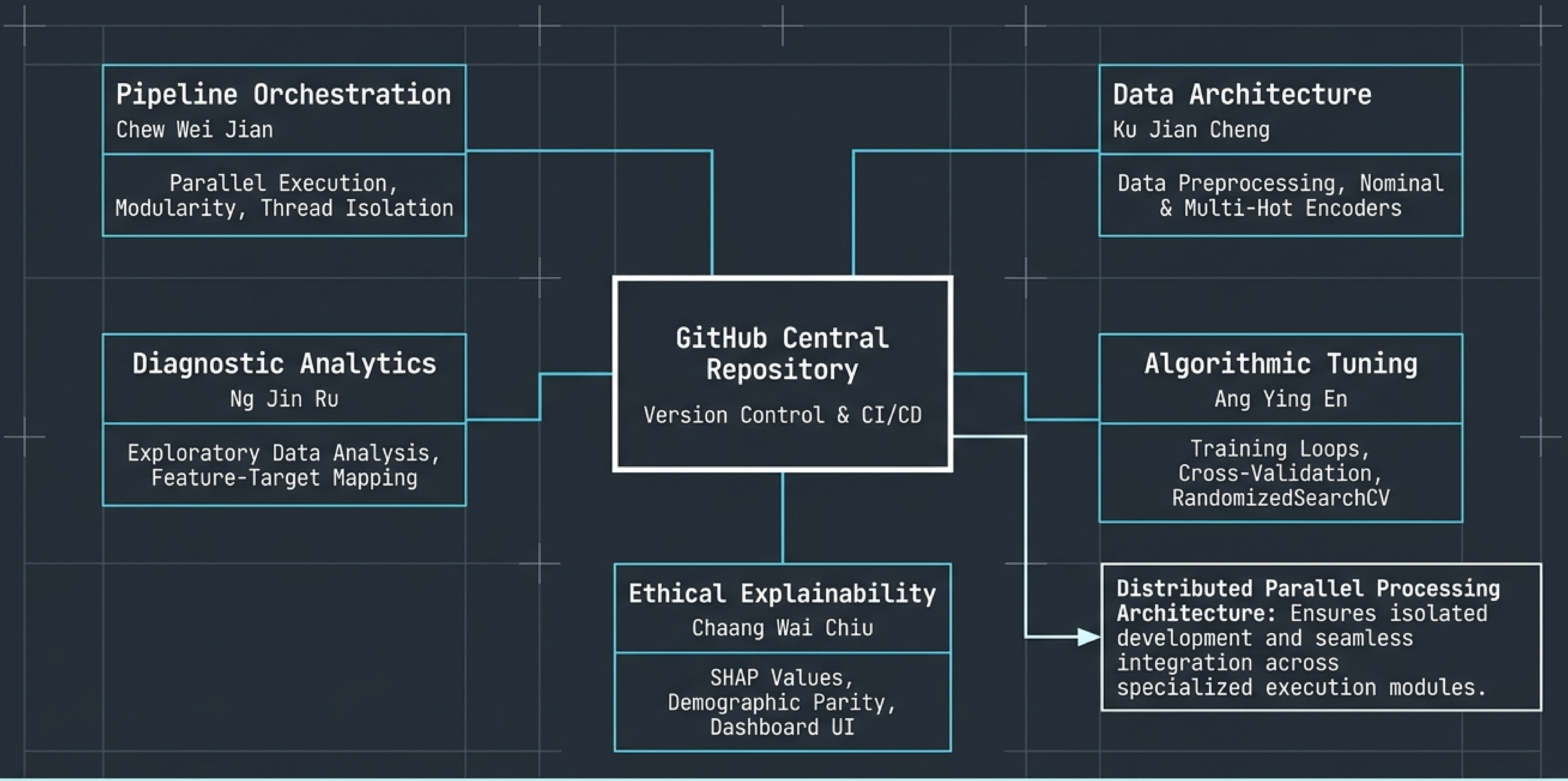

Methodology: Modular Jupyter notebook branches utilizing strict peer-review protocols to guarantee zero code conflicts and complete reproducibility.

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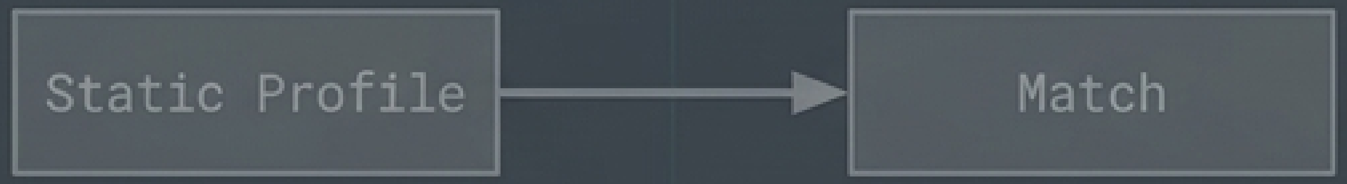
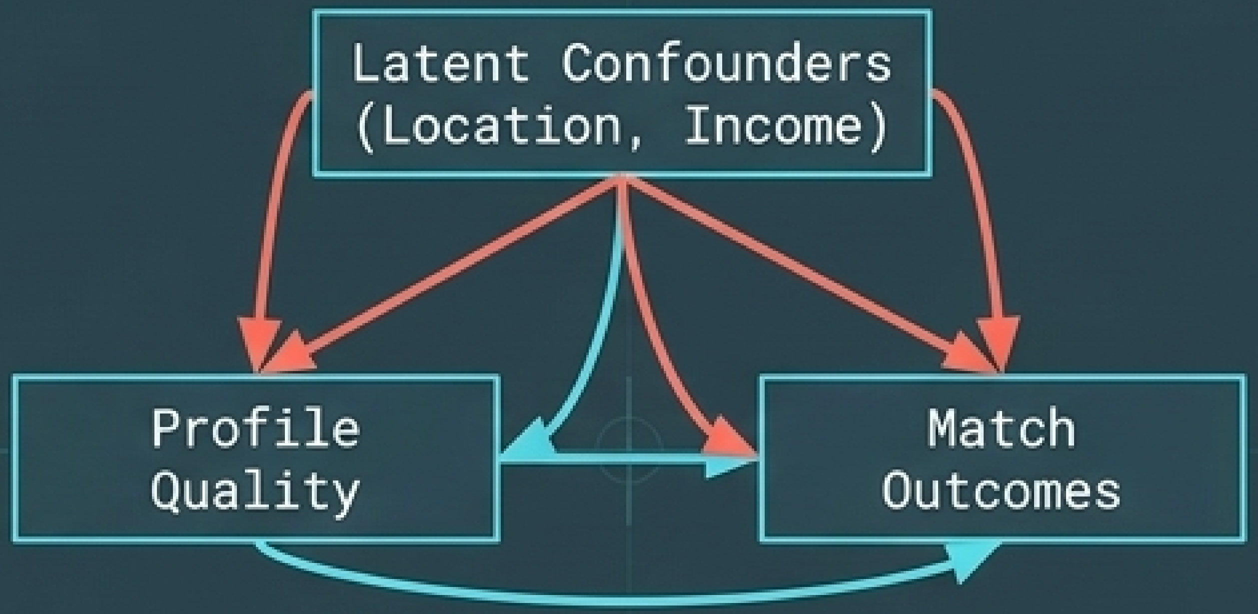
Problem Background & Objective

The Naive Approach



The superficiality of modern matchmaking. Current algorithms optimize for engagement loops and swipe volume, leading to high churn, extreme ghosting rates, and algorithmic hallucination.

The Mathematical Reality



Naive models conflate prediction with decision. They learn spurious backdoor correlations rather than true romantic compatibility.

The Objective: Transition from predictive guessing to prescriptive causal AI. Build a leakage-free causal audit sandbox that isolates true predictive signals from demographic noise using backdoor adjustments.

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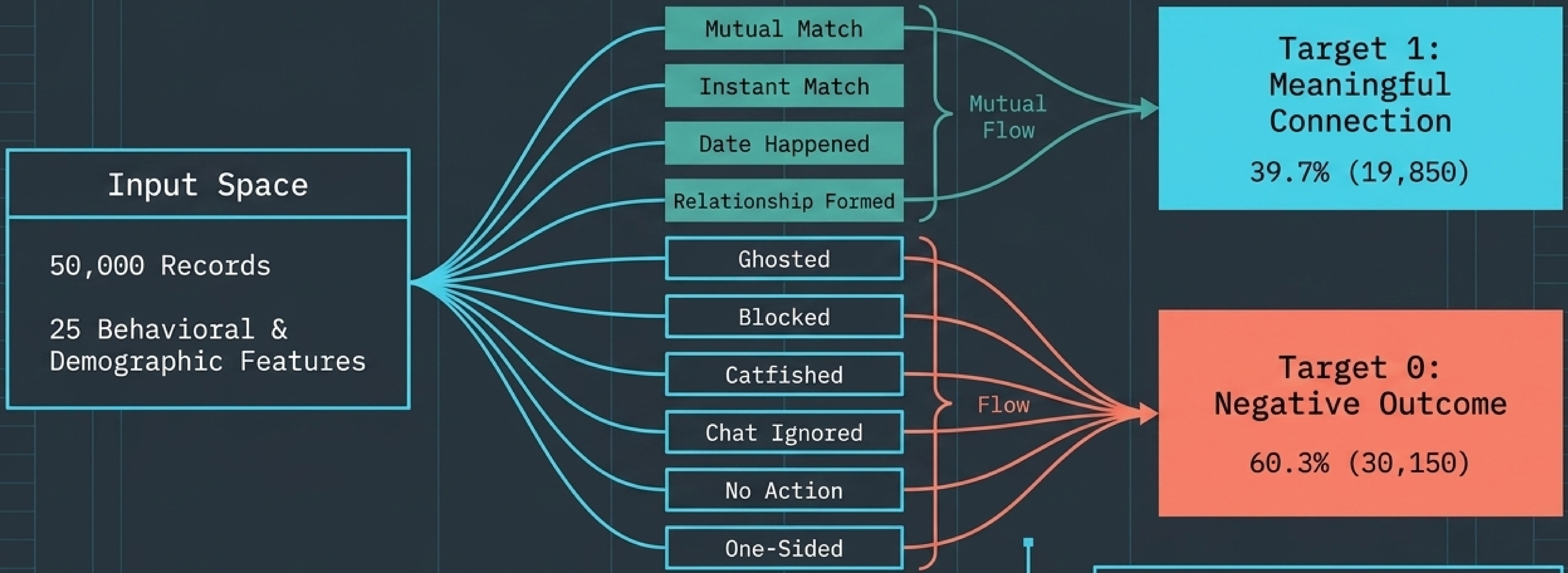
The Dataset & Target Variable



Analytical Takeaway: Simplifies the predictive mathematical objective and yields a stable baseline for SCM causal analysis.

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Exploratory Data Analysis: Categorical Uniformity

Categorical Uniformity Audit

Programmatic Balance:

Perfectly uniform distribution across Gender, Sexual Orientation, and Location variables.

Strategic Advantage:

- Natively enforces a baseline for demographic parity and ethical fairness.
- Mathematically guarantees the loss function cannot be hijacked or dominated by a majority demographic cohort.

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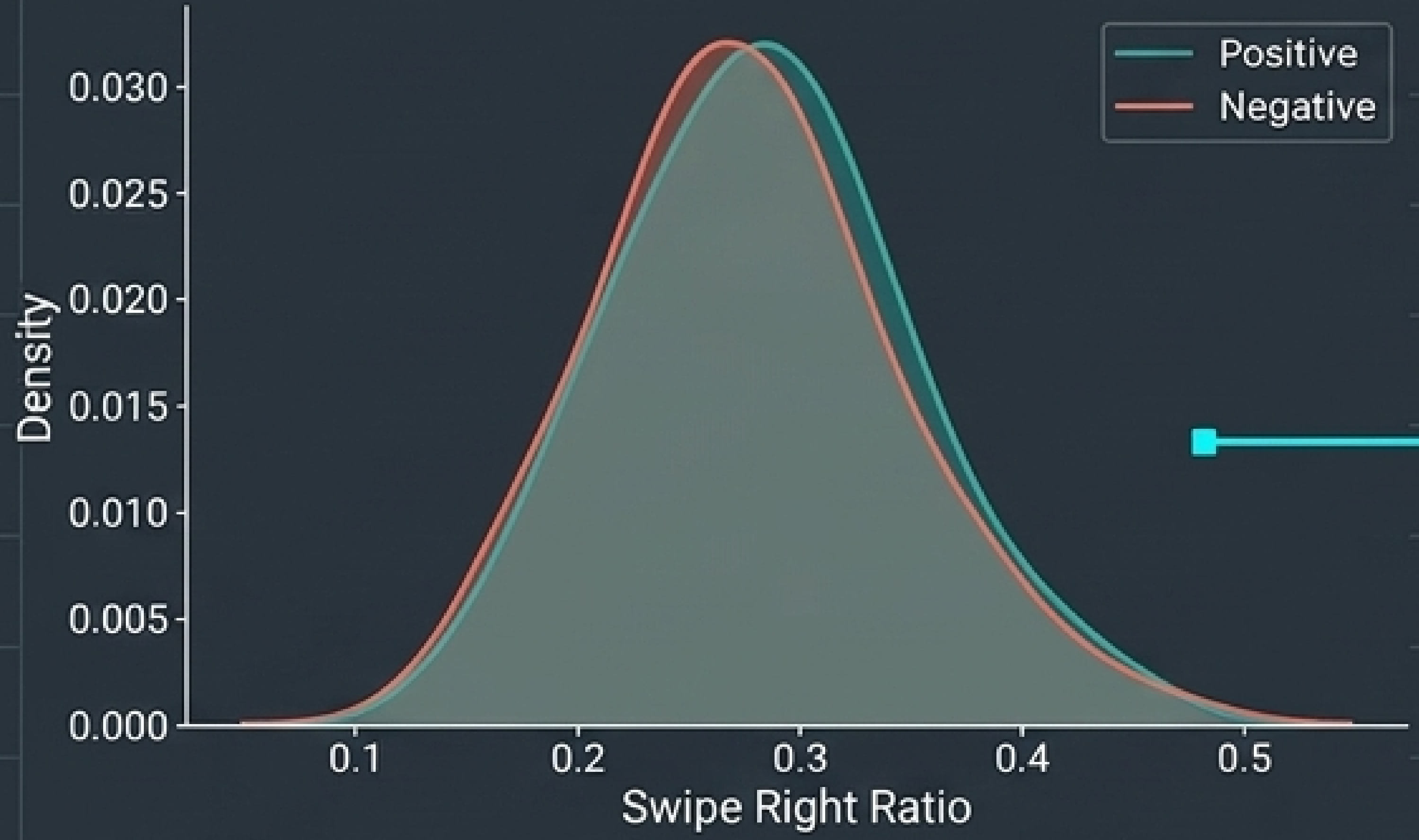
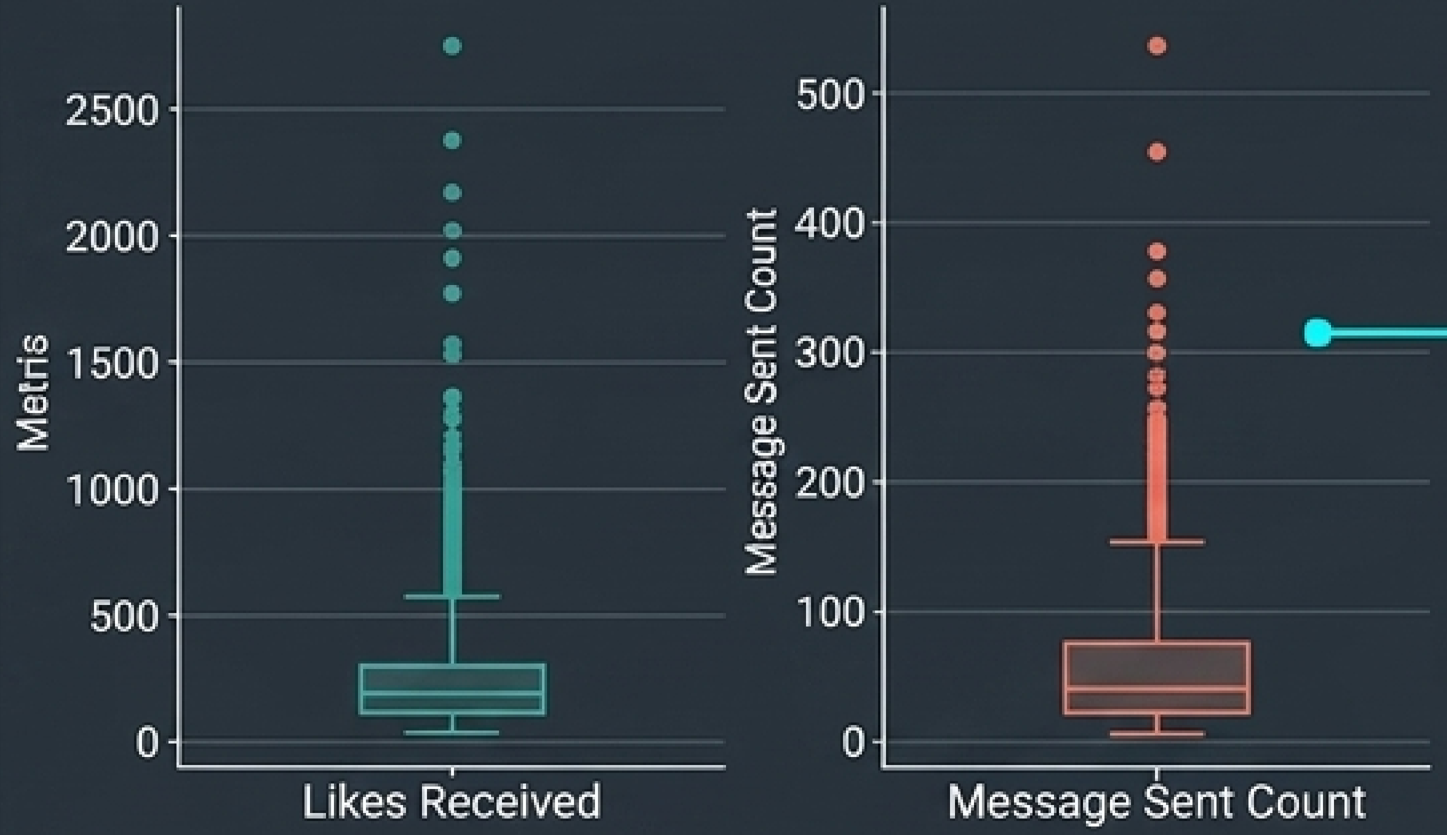
Exploratory Data Analysis: Numerical Density & Outliers

Numerical Density



Zero Linear Separation: Density curves for positive vs. negative targets completely overlap. Univariate linear splits will fail mathematically.

Outliers



Extreme Power-Users: Heavy-tail outliers dominate engagement metrics, rendering mean-variance scaling highly vulnerable.

The Guardrail Decision:

We deliberately abandoned standard scaling and upgraded to the median/IQR-based `RobustScaler`, deferred strictly to post-split to prevent pre-split data leakage.

```
RobustScaler(quantile_range=(25.0, 75.0))
```

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